

# INFO 5602

## Information Visualization

**“Focus Your Audience’s Attention”**

# Agenda:

- “You see with your brain”
- Pre-attentive attributes (i.e. visual cues) signal where to look
- Pre-attentive attributes in text
- Pre-attentive attributes in graphs
  - Size
  - Color
- Group exercise/learning:
  - Morgan Stanley case study
  - J.P. Morgan case study

# We are NOT naturally good at **storytelling** and **presentation structuring**

Presentation = Noise + Signal

- *Clutter/distractions on slides*
- *Stakeholder - irrelevant information included*
- *Stakeholder -irrelevant "design" options*
- *Distracting colors, objects included in presentations*
- ***Adds cognitive load to stakeholders***

Find + Remove

- *Stakeholder-relevant information*
- *Stakeholder-relevant charts*
- *Stakeholder-relevant design options*
- *User-centric design for charts*
- ***Improves effective stakeholder communication***

Find/Create + Boost

Anything you add to this \*blank\* slide adds  
***cognitive load*** to an audience

# Brief Introduction to the concept of *cognitive load*

- Everything you add to a blank screen takes up cognitive load on the part of your audience – i.e. **takes them brain power to process**
- We experience cognitive load anytime we take in information (especially the first time)
- **Cognitive load** is the mental effort that is required to **learn new information**
  - When we ask a computer to do work, we rely on the computer's processing power (CPU, GPU etc.)
  - When we ask our stakeholders/audience to do work we are leveraging their **mental processing power**
  - Any item in your visualization that **adds no value, increases cognitive load**
- For visualizations, identify anything that is not adding informative value – or is not adding enough informative value to make up for its presence – and remove those things to **reduce cognitive load**

# “You see with your brain”

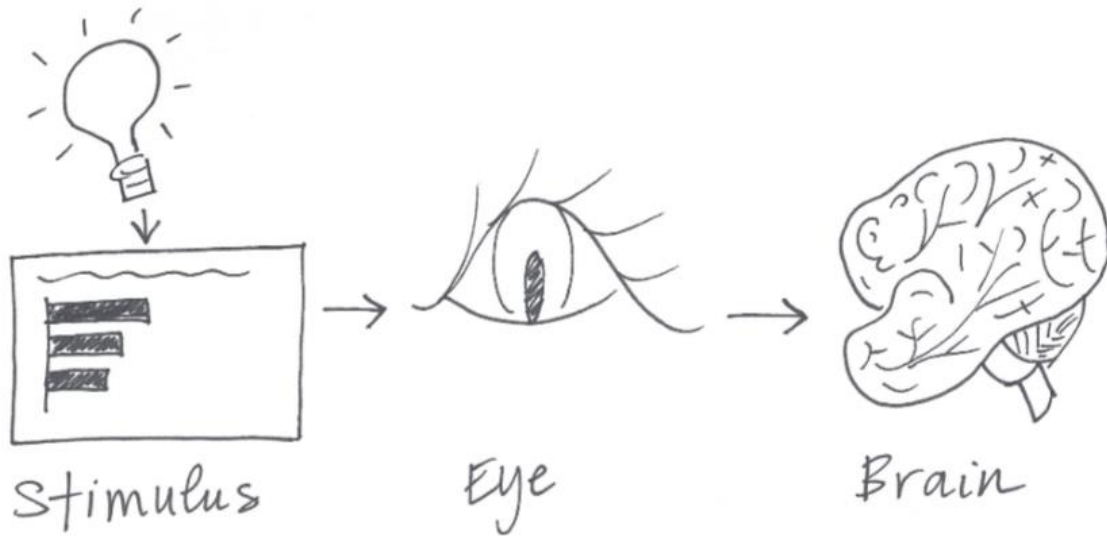


FIGURE 4.1 A simplified picture of how you see

- **Light reflecting off a stimulus travels through the eye and is processed fully by the human brain. We do not fully see with our eyes**
- Most information processing happens inside the human brain – known as *visual perception*
- **Within the brain there are three types of memory important to understand how humans process visual communications:**
  - Iconic memory
  - Short-term memory
  - Long-term memory

# “You see with your brain”: **iconic** memory

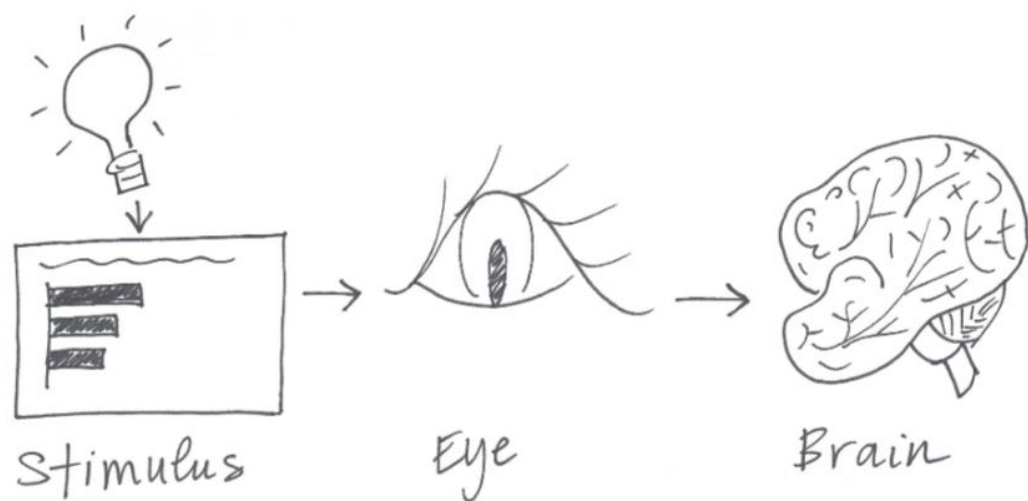


FIGURE 4.1 A simplified picture of how you see

- **Iconic memory is super fast in capturing visual information** and allows processing of visual communications without you fully realizing it
- The ability to pick up and process information around is a service we obtain from the iconic memory of our brains
- Information stays in iconic memory for a fraction of a second before it gets forwarded on to short-term memory.
- The important thing about iconic memory is that it is tuned to a set of pre-attentive attributes
- Pre-attentive attributes are information visualization tools used to boost the *signals* in your presentation

# “You see with your brain”: **short-term** memory

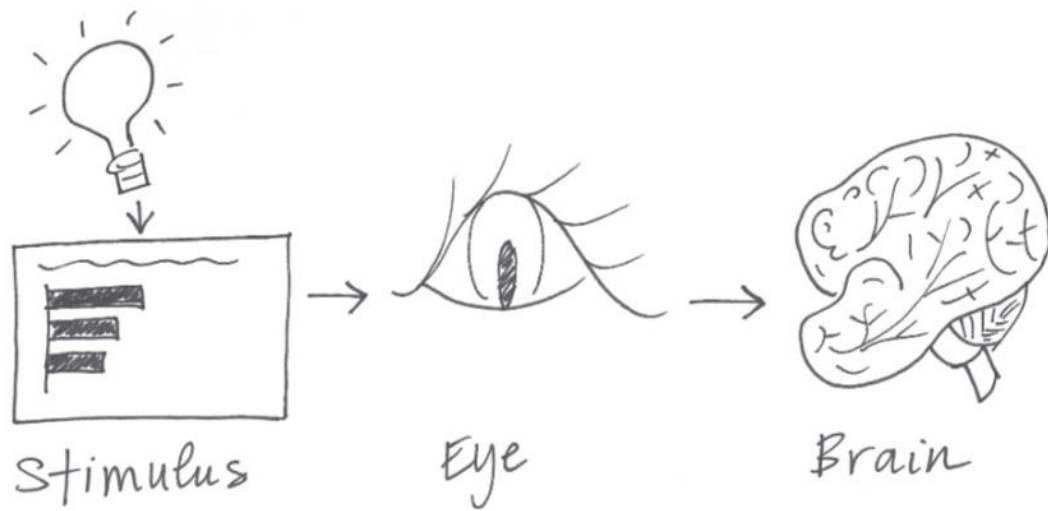


FIGURE 4.1 A simplified picture of how you see

- **Short-term memory receives visual information passed on by iconic memory**
- **Short-term memory is limited** – humans can keep about four chunks of visual information in their short-term memory at a given time – Knafllic (2017)
- So, if you create a dashboard or chart with ten colors, ten different data markers, legend etc, you are making your stakeholders/audiences work very hard to process and retain information (a.k.a cognitive load)
- You need to **keep things simple** when designing your charts and dashboards – so you can fit visual information in the short-term memory of your stakeholders. Recall that humans are:
  - Attention deficient
  - Time-constrained
- Short-term memory passes visual information to **long-term memory**

# “You see with your brain”: **long-term** memory

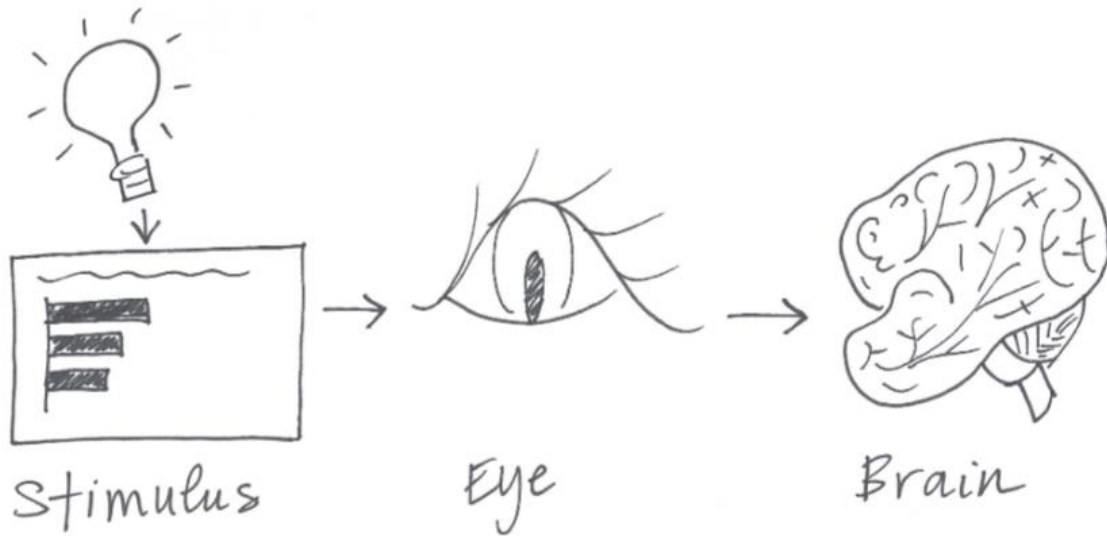


FIGURE 4.1 A simplified picture of how you see

- Long-term memory receives visual information from short-term memory
- **Long-term memory is built up over a lifetime** and is crucial for pattern recognition and general cognitive processing
- Long-term memory is the aggregate of visual and verbal memory, which act differently:
- Long-term memory is very important for pattern recognition and **recall**
- Visual information or images can help audiences recall information stored in long-term memory

# Pre-attentive attributes signal where to look

- Pre-attentive attributes signal where to look
- **Pre-attentive attributes are used to boost the signal in your presentation**
- **Pre-attentive attributes are used to draw your audience's attention quickly to where they need to focus on in your presentation to achieve effective communication**
- **Pre-attentive attributes can be used to create a visual hierarchy of information**

End of presentation